

Proxy Hardening — Test-Type Coverage Status

Revision: 2 **Last modified:** 2026-07-01T13:12:00Z **Status:** In progress — 8 hardening test types PASS with captured live evidence (stress+chaos, DDoS, concurrency/atomicity, memory-soak, **P10 VPN fail-closed full-automation GREEN, race/deadlock, benchmarking/performance**), 1 honest SKIP (security ACL — no autonomous real-ACL-deny topology). P10 GREEN was proven live this round after recreating the 4 dynamic-stack Podman secrets + booting the dynamic stack (tunnel-DOWN ⇒ branded 503 ×3 + NO leak, ×3 deterministic + RED polarity guard). No PASS is claimed without a real qa-results/ evidence path (§11.4.6). **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc reconciling the §11.4.169 test-type-coverage matrix for the proxy hardening workstream. **Companion:** summary [Status_Summary.md](#)

Operator-blocked / pending items (read first — §11.4.45 O(1) surface)

Item	Why blocked / pending	Unblock condition
P10 real-VPN egress-half (SKIP)	The egress half — proving traffic actually EXITS via the tunnel — needs operator gluetun WireGuard credentials (§11.4.21). The fail-closed half is now GREEN.	Operator provides gluetun WireGuard creds; then the real-egress positive path can be captured.
security ACL deny-does-not-leak (SKIP)	No autonomous topology on the base stack produces a real Squid ACL deny (observed deny code 000000 — cannot prove deny nor leak).	A dynamic/ACL-configured topology where a CONNECT to a denied target returns 403/407, so deny-without-leak is provable.

§11.4.169 test-type-coverage matrix (captured-evidence-driven — §11.4.5/ §11.4.69)

Base proxy stack UP on :53128 at capture time. Every PASS below cites a real, non-empty qa-results/ artefact captured live this session.

Test type	Status	Evidence path	Notes
Unit	PENDING	—	Not re-proven hardening-rou Tracked in the
Integration	PENDING	—	Not re-proven hardening-rou
E2E	PENDING	—	Not re-proven hardening-rou
Full- automation (P10 VPN fail- closed)	PASS	qa-results/dynamic/vpn_failclosed/ 20260701T130115Z/verdict.txt (+ req_1..3.body branded pages)	tests/dynamic/ booted dynam ERR_TUNNEL_DOW markers), leak PID unchange polarity guard Egress-half op
Challenges	PENDING	—	Not re-proven hardening-rou
HelixQA	PENDING	—	Not re-proven hardening-rou
DDoS / load- flood	PASS	qa-results/ddos/ proxy_flood_20260701T122320Z/ flood.evidence (+ census.txt)	300/300 reques timeout=0 erro listening after
Security	SKIP	qa-results/security/ proxy_acl_20260701T120913Z/ s1_acl_deny.evidence	Honest SKIP:t real-ACL-deny deny code 0000 (s2_header_hyg (§11.4.3).
Stress + Chaos	PASS	qa-results/stress/ proxy_forward_20260701T120843Z/ stress.evidence · qa-results/chaos/ proxy_restart_20260701T120941Z/ recovery_trace.log	Stress: 110/11 concurrent), v proxy→squid, c recovered cod
Concurrency / atomicity	PASS	qa-results/concurrency/ proxy_concurrency_20260701T122740Z/ concurrency.evidence (+ clients.tsv)	40 simultaneo crosstalk=0 pr talk. OVERALL=P
Race / deadlock	PASS	qa-results/race/control- plane_race_20260701T125739Z.log	go test -race incl. cmd/acl-h

Test type	Status	Evidence path	Notes
			compiler, inter {aclhelper,api EXIT=0, O WAR spawn gorouti (internal/api/ tunnelbreaker. (documented, audited-mutat concurrency/a
Memory	PASS	qa-results/memory/ proxy_soak_20260701T122643Z/ soak.evidence (+ rss_timeseries.tsv)	RSS-bounded 000 B, growth_
Benchmarking / performance	PASS	qa-results/benchmark/ proxy_forward_20260701T130414Z/ latency.txt	tests/benchmark proxy: 200/200 p95=0.088s p99 0.087), throug OVERALL=PASS. F ships zero Go gap (submodu benchmarks).

Honest boundary (§11.4.6)

This matrix reconciles the **hardening** test-type picture proven in this session only. The 8 hardening-focused types (stress+chaos, DDoS, concurrency/atomicity, memory, P10 fail-closed full-automation, race/deadlock, benchmarking) are PASS with captured live qa-results/ evidence. PENDING rows (unit/integration/e2e/Challenges/HelixQA) are not claims the underlying capability is broken — they mean no hardening-round captured evidence exists this session; they are neither PASS nor FAIL and are covered by the broader tests/run-tests.sh suite. The single SKIP (security ACL) is topology-honest (§11.4.3). Two documented follow-ups (store-TOCTOU atomicity gap; control-plane zero Go micro-benchmarks) are §11.4.169 items, not regressions. None of this substitutes for the §11.4.40 full-suite retest before a release tag.